Sr. Scientific Data Engineer, R&D Data Platform

Abbott · United States - Illinois - Abbott Park; United States of America : Remote

Job ID (this archive)	HI-039
Employer	Abbott
Location(s)	United States - Illinois - Abbott Park; United States of America : Remote
Career stage (archive tag)	Experienced
Location group	Other U.S.
Majors (archive tags)	Computer Science, Software Engineering, Data Science
Pay listed	$78,000 – $156,000 per year
Employment type	Full time
Posted / listing date	2026-08-26
Employer requisition	31158977
Recruiting status when checked	Open when checked (Oct 7, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print.
Source page	https://abbott.wd5.myworkdayjobs.com/abbottcareers/job/United-States---Illinois---Abbott-Park/Sr-Scientific-Data-Engineer--R-D-Data-Platform_31158977-1
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Course sheet refs	HI-039

> Captured on 2026-10-07 from the employer's public career site. The posting text below is the employer's own wording, reformatted for printing (site navigation removed). Live listings change and close; check the source page before applying.

Job description

Abbott is a global healthcare leader that helps people live more fully at all stages of life. Our portfolio of life-changing technologies spans the spectrum of healthcare, with leading businesses and products in diagnostics, medical devices, nutritionals and branded generic medicines. Our 115,000 colleagues serve people in more than 160 countries.

JOB DESCRIPTION:

Position Overview:

The Science Office within Abbott Cancer Diagnostics is seeking a Senior Scientific Data Engineer to lead the design and delivery of practical data solutions for cancer research and diagnostic development. This position sits at the intersection of software engineering, scientific data, and applied analysis. You will own significant pieces of our research data platform end to end, building reusable tools and workflows that help researchers organize, validate, discover, transform, analyze, and share complex data. These solutions may include Python packages, data pipelines, APIs, notebooks, workflow utilities, and lightweight web applications. This is not a traditional enterprise data warehousing position. The work centers on heterogeneous research data generated across scientific programs, including genomic, clinical, imaging, laboratory, and experimental data. Successful candidates will combine deep technical skills with an understanding of how quantitative research is conducted, and will be comfortable setting technical direction when a problem is still loosely defined. You will work closely with scientists, data scientists, bioinformaticians, software engineers, and R&D DevOps partners, and will often represent the team in cross-functional technical discussions. The ideal candidate is curious, resourceful, and able to turn ambiguous scientific needs into durable, reusable capabilities, while helping other engineers do the same.

Essential Duties and Responsibilities:

Lead the design and delivery of reusable tools and services for ingesting, validating, transforming, documenting, discovering, and sharing scientific data.

Own one or more platform capability areas end to end, including design, implementation, adoption, operational support, and long-term maintainability.

Develop maintainable solutions using Python and SQL, including software packages, data pipelines, APIs, notebooks, workflow utilities, and lightweight internal applications.

Create approachable, self-service workflows that allow researchers with varying levels of programming experience to prepare and share data consistently.

Partner directly with scientific teams to understand their studies, analytical workflows, data sources, and recurring technical challenges, and translate those needs into a prioritized technical roadmap.

Establish standards and reusable patterns for organizing and harmonizing data from disparate sources, including consistent structures, terminology, variable definitions, and mappings, and drive their adoption across teams.

Design automated data-quality and validation frameworks that identify missing, inconsistent, malformed, or unexpected data before it is used in downstream research.

Improve the documentation, traceability, and discoverability of scientific datasets, including clear descriptions of data content, origin, ownership, processing history, and intended use.

Evaluate AWS services and features for scientific data and analytical workflows. Translate research requirements into technical recommendations and partner with R&D DevOps teams on architecture, deployment patterns, and operational ownership. Develop solutions that use AWS data and analytics capabilities, particularly Amazon S3 and related services such as Athena, Glue, EMR, Lambda, and SageMaker.

Prototype solutions for individual research programs and lead the work of generalizing successful approaches into reusable platform capabilities.

Provide technical leadership on designs that span multiple projects or teams: lead design reviews, document trade-offs and decisions, and align approaches with other engineers and technical leads.

Mentor engineers through code review, pairing, design feedback, and documentation, and raise the overall engineering standard of the team.

Support hands-on preparation and analysis of scientific data when needed to understand a problem, validate a solution, or accelerate a research effort.

Apply quantitative and scientific judgment when evaluating data, analytical requirements, and potential technical solutions.

Use Spark or PySpark when distributed processing is appropriate for large or computationally intensive datasets.

Apply and reinforce sound software-engineering practices, including version control, testing, code review, documentation, dependency management, continuous integration, and reproducible development.

Communicate technical concepts, design decisions, trade-offs, limitations, and project status clearly to technical, scientific, and leadership audiences.

Operate independently within an evolving environment: scope ambiguous problems, sequence the work, make defensible decisions when requirements are incomplete, and keep stakeholders informed.

Minimum Qualifications:

Bachelor’s degree in computer science, data science, engineering, statistics, mathematics, bioinformatics, computational science, or another relevant quantitative discipline.

Five or more years of relevant professional or applied research experience, or three or more years with an advanced degree in a relevant field.

Advanced programming skills in Python.

Strong SQL skills and experience working with structured and semi-structured data.

Demonstrated track record of building reusable, maintainable software that others depend on, rather than one-time scripts or analyses.

Experience designing and delivering several of the following: data pipelines, Python packages, APIs, analytical workflows, notebooks, or internal software tools.

Substantial hands-on experience using AWS for data processing, analytics, scientific computing, or software development.

Sufficient depth in AWS services and architecture to evaluate technical options, justify design recommendations, and define infrastructure requirements with DevOps or cloud-engineering partners.

Experience conducting or supporting quantitative research, such as statistical analysis, machine learning, computational modeling, or another data-intensive research activity.

Experience cleaning, integrating, standardizing, or validating data from multiple sources at meaningful scale.

Fluency with software-development practices such as Git, automated testing, technical documentation, code review, and continuous integration.

Demonstrated ability to investigate ambiguous problems, define an approach, and deliver a working solution with little guidance.

Experience mentoring or providing technical guidance to other engineers, scientists, or analysts.

Strong communication and collaboration skills, particularly when building alignment across scientific and technical disciplines.

Preferred Qualifications:

Advanced degree in a quantitative, computational, or life-science discipline.

Experience working with biomedical, genomic, clinical, proteomic, imaging, laboratory, or other complex scientific data.

Experience supporting research in life sciences, healthcare, diagnostics, or a similarly data-intensive and regulated scientific environment.

Production experience with Spark or PySpark and distributed data processing.

Depth in AWS services such as Athena, Glue, EMR, SageMaker, Lambda, Step Functions, Lake Formation, or related data and analytics technologies.

Experience developing REST APIs or lightweight web applications used by non-engineering audiences.

Experience designing automated validation frameworks, data contracts, reusable data-processing libraries, or researcher-facing workflow tools.

Experience with metadata-management, data-catalog, or data-discovery platforms, such as the AWS Glue Data Catalog, OpenMetadata, or Unity Catalog.

Experience with containerization, continuous integration and deployment, or infrastructure-as-code.

Experience supporting machine-learning workflows or preparing data for model development and evaluation.

Experience working with large files or multimodal datasets, such as sequencing outputs, digital pathology images, clinical records, or experimental measurements.

Familiarity with governance considerations for research data, including access control, de-identification, and the handling of sensitive clinical information.

Experience working within a data mesh, data product, or federated data-ownership model.

What Success Looks Like:

Scientists spend less time manually locating, cleaning, interpreting, and restructuring data.

Research teams have well-supported, documented tools that help them prepare and share data consistently.

Data-quality problems are identified earlier through automated checks, validation at the point of handoff, and clearer documentation.

Approaches you design become shared standards that multiple scientific teams adopt, rather than being rebuilt program by program.

AWS capabilities are selected and applied thoughtfully in partnership with R&D DevOps, with clear ownership boundaries and repeatable deployment patterns.

Other engineers work more effectively because of the patterns, reviews, mentoring, and documentation you contribute.

Scientific data becomes easier to discover, understand, analyze, and reuse across the organization.

The base pay for this position is

$78,000.00 - $156,000.00 In specific locations, the pay range may vary from the range posted.

JOB FAMILY:

Product Development

DIVISION:

ONCO Cancer Diagnostics

LOCATION:

United States of America : Remote

ADDITIONAL LOCATIONS:

WORK SHIFT:

Standard

TRAVEL:

Yes, 10 % of the Time

MEDICAL SURVEILLANCE:

No

SIGNIFICANT WORK ACTIVITIES:

Continuous sitting for prolonged periods (more than 2 consecutive hours in an 8 hour day), Keyboard use (greater or equal to 50% of the workday)

Abbott is an Equal Opportunity Employer of Minorities/Women/Individuals with Disabilities/Protected Veterans.

EEO is the Law link - English: http://webstorage.abbott.com/common/External/EEO_English.pdf

EEO is the Law link - Espanol: http://webstorage.abbott.com/common/External/EEO_Spanish.pdf

Notes for ECS 1100 students

How this connects to your major

Data Science: healthcare data analysis, data engineering, statistics, AI or computational biology; Computer Science: healthcare software, embedded systems, automation or secure computing. These are teaching annotations, not an employer guarantee of eligibility.

Use the degree and experience evidence to distinguish a first job from a later career step.

DS: Named discipline appears in source; consult qualifications for eligibility

CS: Named discipline appears in source; consult qualifications for eligibility

Skills mentioned in the posting

Python, SQL, Git/source control, Machine learning, Statistics, Cloud, Data pipelines/ETL, Software testing, Verification/validation, Genomics/bioinformatics, Communication/teamwork

Course sheet tags

Majors (teaching tags): DS; CS; Career stage (teaching tag): Senior / leadership; Geography: US remote; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Full time.

Source record

Original posting https://abbott.wd5.myworkdayjobs.com/abbottcareers/job/United-States---Illinois---Abbott-Park/Sr-Scientific-Data-Engineer--R-D-Data-Platform_31158977-1

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